



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 3 • STANDARD 6.AT.3

Ratio and Rate Problem Solving

Lesson 3-7 • Teacher Copy

Name: _____

Date: _____

Class: _____

My Notes

Level 2 Standard



TODAY'S OBJECTIVES

Content Objective: I can solve real-world rate problems by finding and using unit rates.

Language Objective: I can explain my solution using the words rate, unit rate, per, and proportion.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Rate

Unit rate

Per

Problem solving

Proportion



Tap any word to see what it means and a picture.

1

A ratio comparing two quantities with different units is a

.

2

A rate for exactly one unit, like \$3 per 1 pound, is a .

3

The word that means 'for each' in a rate, like miles hour, is .

4

Using a plan and math steps to find an answer to a real situation is

.

5

An equation that two ratios are equal is a .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

Team 2 frosts 5 cupcakes every 4 minutes. How did you set up a rate or proportion to find how many cupcakes they frost in a longer time?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

☐ **Rate** — A ratio comparing two amounts with different units, like miles per hour.
Tasa — Una razón que compara dos cantidades con unidades distintas, como millas por hora.

☐ **Unit rate** — A rate for just 1 of something, like cost for 1 item.
Tasa unitaria — Una tasa para solo 1 de algo, como el precio de 1 artículo.

☐ **Per** — For each one. Example: 5 dollars per book.
Por — Por cada uno. Ejemplo: 5 dólares por libro.

☐ **Proportion** — A math sentence saying two ratios are equal. It helps find a missing number.
Proporción — Una oración matemática que dice que dos razones son iguales. Ayuda a hallar un número que falta.

☐ **Problem solving** — Using ratios and rates to find a missing amount.
Resolución de problemas — Usar razones y tasas para encontrar una cantidad que falta.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

The rate is ____ cupcakes per ____ minutes.

La tasa es ____ cupcakes por ____ minutos.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Divide the total cost by the number of pounds to find the cost per 1 pound.

Evidence: Students divide \$18 by 3 to get \$6 per pound and explain the unit rate lets them find the cost of any number of pounds.

Reasoning: This evidence connects the definition of rate to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The rate is ____ cupcakes per ____ minutes.**

La tasa es ____ cupcakes por ____ minutos.

- **To find the total, I ____ the rate.**

Para hallar el total, ____ la tasa.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

Mi afirmación es ____.

- **I know because ____.**

Lo sé porque ____.

- **So ____.**

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.

- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.

- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.

- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Rate
2. **Notes 2:** Unit rate
3. **Notes 3:** Per
4. **Notes 4:** Problem solving
5. **Notes 5:** Proportion
6. **Watch for this mistake:** *A common mistake in Ratio and Rate Problem Solving is dividing in the wrong order when setting up the unit rate — for example, finding "minutes per smoothie" ($6 \div 18$) instead of "smoothies per minute" ($18 \div 6$) — or using addition instead of multiplication to scale the unit rate, like adding $\$0.75 + 10$ instead of multiplying $\$0.75 \times 10$. Before you submit, check: does your unit rate match the units the question is asking for, and did you multiply (not add) to scale it to the new amount?*
7. **Try It 1:** A. 28 carrots — *Unit rate: $8 \text{ carrots} \div 2 \text{ minutes} = 4 \text{ carrots per minute}$. In 7 minutes: $4 \times 7 = 28 \text{ carrots}$.*
8. **Try It 2:** A. 42 ounces — *Unit rate: $18 \text{ ounces} \div 3 \text{ loaves} = 6 \text{ ounces per loaf}$. For 7 loaves: $6 \times 7 = 42 \text{ ounces}$. The equivalent ratio $18/3 = 42/7$ confirms it.*